Fear of the invader: a predatory invader may provoke stronger non-consumptive effects on a keystone native prey than a co-evolved native predator

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Non-consumptive effects (NCEs) of predators on prey encompass costly antipredator behaviours such as increased refuge use and reduced foraging, which can influence prey population dynamics and community structure, particularly when the prey themselves are keystone species. In Aotearoa-New Zealand, the native freshwater crayfish kōura (*Paranephrops planifrons*) is a keystone species, an ecosystem engineer with a critical functional role in sediment dynamics, nutrient cycling and benthic community structure. Kōura are preyed upon by both the co-evolved native longfin eel (*Anguilla dieffenbachii*) and invasive brown bullhead catfish (*Ameiurus nebulosus*), a species introduced less than two centuries ago. Under laboratory conditions, we tested whether visual and chemical cues from the two predators induced different changes in kōura refuge-seeking and feeding behaviour. In a mesocosm habitat experiment, predator presence produced a NCE on refuge use, with kōura exposed to catfish cues showing significantly greater refuge-seeking behaviour than those exposed to eel cues. This is consistent with a generalised overcautious fear response triggered by unfamiliar predator cues rather than classic prey naivety. Neither predator treatment differed significantly from the control, however the catfish–eel contrast was robust and biologically meaningful. In the kōura feeding experiment, exposure to catfish and eel kairomones had no clear effect on feeding rate. Our study indicates catfish incursions may provoke a greater NCE on kōura refuge use than resident eels. As catfish range expansion continues, these NCEs represent an ecologically significant and underappreciated threat to kōura populations, compounding the direct predation pressure already documented for this invader.

Table of contents

1 Introduction **2**

2 Materials and Methods **3**
- 2.1 Animal collection and preparation . . . 3
- 2.2 Refuge experiment . . . 4
- 2.3 Feeding experiment . . . 5

3 Results **6**
- 3.1 Refuge experiment . . . 6
- 3.2 Feeding experiment . . . 7

4 Discussion **7**
- 4.1 Conclusion . . . 9

Acknowledgements **10**

Authors' contributions **10**

Ethical approval **10**

Declarations **10**

Data availability statement **10**

Figures **11**

Supplementary information **14**

1 Introduction

Biological invasions can have profound impacts on invaded ecosystems, particularly through the introduction of novel predators that prey upon native species lacking an evolutionary history with them (Doherty et al., 2016). Native prey are often naïve to these predators, making them especially vulnerable to direct predation (Gamradt & Kats, 1996; Smith et al., 2008). In addition to these consumptive effects, native prey may exhibit heightened fear responses to invasive predators, leading to the development of anti-predator behaviours such as increased refuge use and avoidance of open habitats (Kindinger & Albins, 2017). These behavioural changes, termed non-consumptive effects (NCEs), can substantially influence population dynamics by reducing foraging activity, growth, and reproductive output (Preisser et al., 2005). In contrast, native predators generally elicit different responses because prey and

predators have co-evolved, resulting in behavioural and physiological adaptations that stabilise predator–prey interactions within local ecosystems (Abrams, 2000).

Freshwater crayfish often live in turbid flowing water or in the dark depths of lakes, therefore they often rely primarily on chemical cues (kairomones) rather than visual cues to detect and respond to predatory fish (BLAKE & HART, 1993; Brönmark & Hansson, 2012; Wood & Moore, 2020b). The kairomones carry signals via excreta, mucus, or skin, that crayfish use to assess predator risk, responding more strongly to predators that have consumed conspecifics and to alarm cues released when prey are injured or killed (Beattie & Moore, 2018; Gherardi et al., 2011). Exposure to kairomones from predatory fish can generate NCEs, including reduced growth and reduced activity levels in crayfish (ADAMS, 2007; Ramberg-Pihl & Yurewicz, 2020), and even non-predatory fish can cause these behaviour changes, likely through heightened generalised awareness of risk (Nyström, 2005). However, the strength of these responses depends on prior experience as wild-caught individual crayfish show stronger antipredator responses than farmed conspecifics naive to predators (Martin, 2014), indicating that both individual experiences and evolutionary history shape kairomone detection.

In Aotearoa-New Zealand (hereafter Aotearoa), the endemic northern freshwater crayfish *Paranephrops planifrons* White, 1842 commonly known as kōura is a 'culturally treasured' taonga species and a valued food resource for Māori the indigenous people of Aotearoa (Kusabs & Quinn, 2009). Ecologically, kōura play a critical role in freshwater ecosystems as ecosystem engineers through their effects on sediment dynamics, nutrient cycling, and benthic community structure, and have been described as keystone species in many freshwater systems (Collier et al., 1997; Parkyn et al., 2001; Usio & Townsend, 2004). Kōura have a long co-evolutionary history with the native longfin eel (*Anguilla dieffenbachii*) (hereafter eel) and shortfin eel (*A. australis*) (L. A. Brown, 2009; MacNeil et al., 2025) and have been shown to respond more strongly to eel kairomones than to those of introduced non-native brown trout (*Salmo trutta*). This was even when kōura were sourced from streams where both eel and trout predators were present, suggesting that co-evolutionary history, rather than individual experience alone, shaped kōura predator recognition (Shave et al., 1994). Currently kōura are threatened by multiple interacting stressors, including climate change, water-quality degradation and invasive species, contributing to widespread population declines (Kusabs et al., n.d.; Lee et al., 2025). One of the invasive species that threatens kōura is the non-native brown bullhead catfish (*Ameiurus nebulosus*) (hereafter catfish) which was introduced to Aotearoa in 1878 and has since spread through various waterways on the North Island (Barnes & Hicks, 2003; Dedual, 2019; Lee et al., 2025; McDowall, 1990). The brown bullhead catfish is highly tolerant of a wide range of environmental conditions (Lee et al., 2025; Scott & Crossman, 1973) and, due to its broad gape and opportunistic feeding behaviour, is capable of consuming a wide variety of native aquatic organisms (Dedual, 2019; Scott & Crossman, 1973). The recent and ongoing spread of brown bullhead catfish into North Island waterways therefore represents a novel predation threat to which kōura may lack adequate behavioural or sensory adaptations.

In this study, controlled laboratory experiments were used to test whether predator presence and associated chemical cues induced changes in refuge-seeking and feeding behaviour, thereby

providing insight into the behavioural mechanisms underlying NCEs of kōura responses to native and invasive predation threats. It is expected that kōura would show more shelter-seeking behaviour when presented with eels compared to catfish, due to naivety of the kōura towards a novel predator such as catfish (Carthey & Banks, 2014). However, an alternative prediction, consistent with fear generalisation theory, is that kōura may instead exhibit a heightened, overcautious response to the unfamiliar catfish cues, overresponding relative to the actual predation risk (Sih et al., 2023). These two hypotheses generate opposite predictions, and the present study was designed to test which better describes kōura behavioural responses to native and invasive predator cues. Further, it is expected that light regime would have an influence on shelter seeking behaviour of kōura, which like many crayfish, are most active nocturnally (Devcich, 1979; Johnson et al., 2014). It was also predicted that kōura feeding would be more inhibited when exposed to eel kairomones than to those of the novel catfish, reflecting the same co-evolutionary recognition expected to drive greater refuge use in response to eels. The overall aim of the refuge and feeding behaviour studies are to contribute to understanding the behavioural mechanisms underlying kōura vulnerability to invasive catfish and how these may differ from and be additional to, those already manifested in the presence of native predators such as eels.

2 Materials and Methods

2.1 Animal collection and preparation

In mid-August 2025, kōura were sourced from two catfish-free lakes, (Ōkāreka and Tikitapu (Blue Lake)) in the Rotorua Te Awara Lakes area, Bay of Plenty region of the North Island. In late September 2025, catfish and eels were sourced from Lake Rotoroa in Hamilton in the Waikato region of the North Island. Experiments were conducted at the Aquatic Research Centre of the University of Waikato. All three species were held separately in holding tanks and acclimated for a week in dechlorinated tap water under the following controlled regime: a 10:14 light: dark cycle reflecting the correct seasonal day-length at a stable temperature of 15.6 °C, pH 8.4 and 10 mg l^{1} dissolved oxygen, the latter parameters reflecting typical physiochemical condition in source lakes. Kōura and fish were fed every other day on Hikari Tropical carnivore fish food pellets (animal protein 47%). Because kōura are taonga (culturally important) species, permission had to be sought from relevant iwi (Māori tribal unit), as well as relevant government regulator to source a limited number of individuals. Consequently, the experiments were carefully adapted, particularly the feeding study to maximize the use of this finite sample. Similarly, eels, also regarded as taonga, along with individual catfish, were reused throughout the study, respecting ethical and welfare considerations, since their presence alone fulfilled the experimental requirements.

2.2 Refuge experiment

Prior to the experiments, all animals were individually weighed and measured. Only male kōura were used, as male crustaceans are usually the most bold and aggressive (Figler et al., 2005; Stein & Magnuson, 1976) and restricting test animals to a single sex, allowed a smaller overall number of kōura to be used. Kōura had a mean (± SD) orbital carapace length (OCL) of 26.0 mm (± 3.08) (18.35-32.95 mm range) and a mean wet weight of 11.7 g (± 4.25) (3–23 g range). Catfish ranged in length from 260-267 mm and 2.34–3.16 kg wet weight and eels ranged in wet weight from 2.50-3.52 kg, both these size ranges being able to consume adult kōura (Barnes & Hicks, 2003; Hicks, 1997; Ryan, 2009).

Experiments were conducted in 38 L glass tanks (600 × 200 × 220 mm) with each tank having three sides (apart from front of tank) blacked out to block visual cues from neighbouring tanks. Each tank was divided into two compartments by a 0.5 mm steel mesh divider, allowing visual and chemical cues between compartments, while preventing any physical contact between fish and kōura on either side of barrier (Figure 1). All tanks were filled with ~26 L dechlorinated tap water and had identical environmental regimes as previously detailed for acclimatisation tanks, with water flow provided by an air stone lodged at the top of the barrier in each tank. Each tank housed a single kōura on one side of the barrier and either a single predator (catfish or eel) or no predator (control) on the other side. A halved terracotta pot (90 mm diameter × 90 mm depth) was positioned on the kōura side of the tank to serve as a potential refuge. To control for potential tank location or compartment side effects, the position of predator treatments within the aquaria was systematically rotated between experimental rounds. The experiment was repeated across five rounds, resulting in 11 catfish replicates, 11 eel replicates, and 10 control replicates and was first carried out in the light (~400 lux; n = 32) before being repeated in the dark (~70 lux; n = 32). Observations were made in the dark using red lighting positioned centrally above the tanks to minimize disturbance, as many aquatic species have limited sensitivity to red wavelengths and are unlikely to perceive this light. Following each individual experimental run, each tank was thoroughly cleaned and refilled with dechlorinated tap water before reuse. Each individual kōura was only exposed once to a fish predator and there were no mortalities of kōura or fish during experiments. All experiments were conducted within a three-week period during October 2025.

At the beginning of each experiment, an individual kōura was placed in the middle of the tank compartment containing the refuge and observations of each kōura location were recorded at 21 time intervals within a 5 hour period (the intervals being: 0, 5, 10, 20, 30, 45, 60, 61, 65, 70, 80, 90, 120, 150, 210, 240, 245, 260, 275, 290, and 300 minutes). The 0-minute recording was the individual's initial immediate location after its addition to the tank. Kōura location was recorded as either refuge (the animal was using the refuge, being inside it or immediately behind it and touching it) or non-refuge/ in the open (any other location in the tank where the animal was not in the refuge or touching the refuge). The fish predator was introduced into the tank on the other side of the barrier from the kōura compartment, after 60 minutes and removed after 240 minutes. The control tanks had no fish added but at 60 minutes and

again at 240 minutes, the water in the 'predator' compartment was mechanically disturbed by agitating the water with a small hand net for approximately 30 seconds. This was to simulate the disturbance time associated with adding and removing fish in the other treatments with the same size hand net. At the end of each experiment, the kōura was removed after the final 300-minute observation.

To analyse kōura refuge use, a beta-binomial generalized linear mixed-effects model (GLMM) was fitted using the glmmTMB package in R (Brooks et al., 2017; R Core Team, 2025). The response variable was the number of observations where kōura were using a refuge out of total observations per experiment round, treatment (catfish, eel, control), light condition (light, dark), period (before, during, after fish addition / mechanical disturbance) included as fixed effects. A beta-binomial distribution was used to account for overdispersion in the binomial proportion data. Model selection was conducted by comparing candidate models differing in fixed effect structure and random effects inclusion using AIC, with the model with the lowest AIC selected as optimal. The need for a tank-level and round-level random intercept was evaluated by comparing AIC between models with and without this term. Model fit of the final model was assessed using DHARMa simulation-based residuals, including tests for overdispersion and zero-inflation (Hartig et al., 2024). Marginal means and pairwise contrasts between factor levels were obtained using the emmeans package (Lenth et al., 2026). Spearman rank correlations were also used to investigate potential relationships between kōura body size (OCL) and overall refuge use (all time intervals combined) in the three treatments.

2.3 Feeding experiment

Experiments were conducted in glass tanks (320 × 170 × 170 mm), containing 5 L dechlorinated tap water and a halved terracotta pot (90 mm diameter × 90 mm depth). Catfish and eels had been held at the same density in separate identical holding tanks for a week prior to experiment. Fish exudates were collected by placing four fish (catfish or eel) in a 10 L bucket filled with water from the holding tanks for 15 minutes to concentrate the kairomones in the water. For eel and catfish treatments, 1 L from the buckets was added to each tank already holding 5 L of tap water, and for control an additional 1 L of tap water was added. Next, 10 Hikari Tropical carnivore pellets were added to each tank. Trial experiments showed these pellets remained intact for over 24 hours in tap water. In three rounds, a total of 48 kōura (46 male, 2 female) were weighed and measured. Their mean (± SD) wet weight of 15.65 g (± 8.84), ranging from 6.00 to 44.00 g, and the mean (± SD) OCL was 27.84 mm (± 4.72), with a range of 20.48 to 41.15 mm. Each kōura was then placed individually in a tank. For each round kōura were starved for different periods (either 0, 3 or 4 days) prior to the feeding experiment to also investigate how starvation period influenced feeding motivation. After 0 days starvation, 16 kōura (14 male and 2 female) were exposed to eel water (n = 5), catfish water (n = 6) or control water (n = 5). After 3 days starvation, 16 kōura (male only) were exposed to eel water (n = 6), catfish water (n = 5) or control water (n = 5). After 4 days starvation, 16 kōura (male only) were exposed to eel water (n = 5), catfish water (n = 5) or control water (n = 6). After

24 hours, when each experiment was terminated, the number of pellets eaten (to the nearest quarter pellet) was counted and the feeding rate (pellets eaten per gram of wet weight (WW) of kōura) calculated. As stated previously, the feeding study was adapted to make the best use of a finite number of collected taonga animals. Kōura feeding rate was analysed using a linear model with pellets consumed per gram body weight as the response, with treatment (catfish, eel, control) and starvation period (0, 3, 4) as fixed effects. Because each starvation period was conducted in a separate experimental round, starvation period and round were fully confounded by design, precluding the inclusion of round as a random effect. Overall significance of fixed effects was assessed using Type III ANOVA [car package; Fox & Weisberg (2018)], and model assumptions were verified using residual plots and testing normality of residuals using the Shapiro-Wilk test. Pairwise comparisons between treatment and starvation level combinations were conducted using estimated marginal means.

3 Results

3.1 Refuge experiment

The proportion of time kōura spent in refuge varied across treatments, experimental periods, and light conditions (Figure 3, Table 1). A beta-binomial GLMM without a random intercept was the best-supported model based on the lowest AIC, with full model coefficients reported in Table 2. Refuge use by kōura was consistently higher when exposed to catfish than eel (p = 0.01, Figure 3, Table 3). However, there were no significant differences in kōura refuge use between either fish treatment and the control (both p = for catfish and control and eel and control; Figure 3). Under light conditions, the pattern of refuge use in the control treatment was more similar to the catfish treatment, whereas under dark conditions it more resembled the eel treatment (Figure 2). Light regime had a significant effect on refuge use (p =), with greater refuge use under light compared to dark conditions (Figure 3 and Figure 2). Regarding experimental periods, refuge use was significantly lower in the period just before fish were introduced (or first 30-second mechanical disturbance initiated) compared to period just after the fish had been removed (or second mechanical disturbance ceased) (p = 0.007; Figure 3 and Figure 2). No significant differences were found between the before and during periods, or between the during and after periods. There was also no relationship between kōura OCL and refuge use in any of the three treatments (rs = -0.09, 0.09 and 0.17 for catfish, eel and control treatments respectively, all NS).

For all treatments, we observed individual kōura within the laboratory population appeared highly variable in their movement around tanks and use of refuges, even in the time intervals comprising the first 60 minutes of the experiment, before fish were introduced. The coefficient of variations (CVs) for kōura refuge use in the light and the dark before fish introduction were similarly high, being 60.4 % and 58.2 % (both n = 32) respectively, confirming high baseline variation in kōura refuge use, regardless of treatment type. Despite this inherent variation in individual kōura behaviour, clear patterns in kōura refuge use were evident over the course

of the experiment. Under light conditions, the control showed the largest overall increase in refuge use (4040% to 8282%), driven primarily by a sharp rise in the after period, while the catfish treatment showed a more sustained increase across all periods (5252% to 7373%) and the eel treatment the smallest increase (3535% to 5353%; Figure 2). Under dark conditions, the pattern shifted, with the catfish treatment showing the greatest increase (3939% to 7070%), while the eel treatment showed a modest and non-sustained increase (2727% to 4141% during exposure, returning to 3636% after), and the control remained comparatively low throughout (3737% to 4747%).

3.2 Feeding experiment

While the linear model showed starvation period had a significant effect on feeding rate ($p = 4 \times 10^{-4}$, Figure 4; Table 4), there was no significant difference in feeding rate between catfish, eel, and control treatments ($p = 0.968$). There was a marginal interaction between treatment and starvation period ($p = 0.082$), with feeding rate in the control treatment appearing to increase with starvation period in a pattern not evident in the fish treatments. After 4 days starvation, the mean (±SE) kōura feeding rate (pellets g WW ^{1}) for the control was over two and a half times that of the catfish and eel treatments (0.148 ±0.050 pellets g WW ^{1} compared to 0.059 ± 0.017 and 0.052 ± 0.028 pellets g WW ^{1} respectively).

4 Discussion

Predatory fish presence can generate a NCE on kōura by increasing the amount of time they spend in refuges. This can reduce the time available for foraging which may lead to reduced growth and reproduction, ultimately impacting the prey species at the population level (Preisser et al. (2005); Sheriff et al. (2020)). Although it was notable that refuge use in all treatments before fish were added, was similar and highly variable, our experiments nevertheless revealed clear differences between kōura responses to catfish and eels, with the invader consistently provoking greater refuge-use than the native. However, the control treatment, which had kōura exposed only to two 30 second periods of water disturbance during the same period that kōura were exposed to fish predators for three hours, also elicited increasing refuge use over time, particularly under light conditions. This is unsurprising given that kōura are nocturnally adapted, with the need to remain hidden during daylight to avoid visual predators such as birds (Devcich, 1979), but does indicate our interpretation of apparent NCEs need to be treated with caution, as the experimental set up was highly artificial and despite acclimatisation efforts, all kōura responses need to be judged in that context. Despite this, the same pattern of catfish presence causing greater refuge use than eel, in the dark, as well as the light does indicate we were witnessing real differences in NCEs.

When exposed to an unfamiliar, novel predator cue, kōura significantly increased their refuge use, indicating a strong antipredator response. For naïve prey, unfamiliar cues could trigger a

precautionary response because the consequences of ignoring a potentially real predation threat could be fatal, even when, in reality, the actual predation risk is absent (Dawkins & Krebs, 1979; Lima & Dill, 1990). Such generalised responses to novel stimuli tend to be overcautious and energetically costly and are characteristic of prey with no evolutionary history with the encountered predator (Sih et al., 2010). This is consistent with the pattern observed in this study as, kōura demonstrated greater refuge-seeking behaviour when exposed to the non-native catfish than to native longfin eels, a difference that likely reflects their evolutionary history with eels but not catfish. The unfamiliar and unpredictable cues from catfish may therefore trigger a heightened precautionary response relative to the more familiar eel cues (Ehlman et al., 2019; Sih et al., 2023). While refuge use in the control group was intermediate and did not differ significantly from either predator treatment, the consistent difference between catfish and eel treatments, observed under light and dark conditions, supports a genuine and manifest difference in NCEs generated by native and non-native predator presence. The energetic cost of this elevated refuge use is itself a meaningful NCE, but the behaviour may simultaneously confer protection against direct predation. Experimental evidence that shelter reduces catfish predation on crayfish (ADAMS, 2007; Zhao et al., 2024) raises the possibility that providing or restoring refuge habitat in catfish-invaded lakes could be a practical management tool to reduce kōura vulnerability, though direct experimental testing in field conditions would be needed to confirm this.

In contrast to catfish exposure, kōura reduced their refuge use when exposed to eels. This likely reflects the co-evolutionary history the two species share, as kōura are able to recognise eel chemical and visual cues and respond in an appropriate rather than precautionary way (Shave et al., 1994). Instead of hiding, kōura may rely on contextual information, such as predator size, movement patterns, or chemical concentrations, to modulate their response (Wood & Moore, 2020a). This pattern was most pronounced in the dark conditions, where all three treatments began similarly but changed progressively over the course of the experiment with catfish treatment steadily increasing in refuge use, while eel treatment stayed alongside the control, declining to baseline levels by the end of the during period. This suggests that under conditions most relevant to kōura natural activity, chemical recognition of the familiar eel predator is sufficient to enable a calibrated, low-cost response, while the unfamiliar catfish continues to elicit sustained precautionary refuge use (Shave et al., 1994; Wood & Moore, 2020b). Beyond predator recognition, one possibility is that our artificial refuges offer limited protection against eels, which can pursue prey into very confined spaces, making active avoidance and chela display a more effective strategy than sheltering (Shave et al., 1994).

The feeding experiment which unlike the refuge-use experiment, relied purely on non-visual cues, did not demonstrate a clear fish predator NCE on kōura. This may have been due to the concentration of predator kairomones not being high enough to exceed detection thresholds (G. E. Brown et al., 2004), the relatively short pre-experiment starvation period for kōura and the restricted time frame of the experiment or a combination of all three factors. Kōura can readily detect necrotic fish tissue and readily leave refuges to scavenge on catfish, eel, and marine fish carcasses with little discrimination between fish species, indicating that chemosensory responses in kōura may depend on cue type, context, and concentration rather than predator identity

alone (MacNeil et al., 2025). The concentration of live fish exudates in our experiment may not reflect natural conditions where kōura and fish predators coexist, and higher concentrations may be required to elicit a detectable feeding suppression response (Achtymichuk et al., 2025). In addition, many crayfishes show a stronger response to alarm cues of conspecifics (Gherardi et al., 2011; Ramberg-Pihl & Yurewicz, 2020; Stebbing et al., 2010), and to kairomones from predators on a conspecifics diet (Beattie & Moore, 2018). Fish predators in our experiments were kept for a prolonged period on the same diet as kōura, which may have further diminished or blunted the threatening nature of signals received from kairomones. In addition, the pre-experiment starvation period for kōura may have been insufficient to elicit a very strong feeding response, as crayfish such as kōura are opportunistic predators which are necessarily adapted to going for prolonged periods without feeding (Hazlett, 2003). The length of time since feeding can influence foraging behaviour, with previous studies indicating that crayfish will be more active foragers after being starved for at least 3 days (Hazlett, 2003). In the current study, the longest starvation period was 4 days and at this point there was a clear discrepancy between the control feeding rate and the much lower feeding rates in the two fish treatments. Allowing for ethics and welfare concerns, if we had extended the starvation period, this may have elicited a stronger and prolonged response. Another potential issue could be as feeding in this study was measured as a single endpoint after 24 hours, this provided no information on whether kōura initially suppressed feeding after immediate exposure to kairomones, before eventually resuming normal feeding activities (Wright et al., 2010). Further experiments addressing these various issues may clarify whether these non-visual cues manifest as NCE's on kōura feeding.

4.1 Conclusion

This study provides experimental evidence that the invasive brown bullhead catfish exerts stronger NCEs on the native freshwater crayfish kōura, evidenced by significantly greater refuge-seeking behaviour than the co-evolved native longfin eel. The heightened response to the novel catfish could reflect a generalised antipredator response triggered by unfamiliar chemical and visual cues, consistent with a fear generalisation framework. In contrast, the reduced response to eels suggests that co-evolutionary experience may have enabled kōura to make more calibrated risk assessments and adjusted refuge use accordingly. Further comparative studies of kōura sourced from catfish invaded freshwaters as well as catfish absent areas would be worthwhile, to assess if catfish 'experienced' kōura react differently than catfish-naive kōura, when presented with the same catfish predator threat. If this was the case and despite catfish being only present in Aotearoa for two centuries, it could present the possibility of 'teaching' naive kōura to adjust their behaviour in an appropriate and proportionate way to catfish presence, by exposing selected naive populations to catfish under controlled laboratory conditions, before an actual catfish invasion event occurs. Although we found no significant NCE of fish exudates on feeding behaviour, our study suggests further investigations of such non-visual cues are warranted in this area. Because kōura are keystone species and ecosystem engineers, even a small increase in refuge use driven by catfish presence could have cascading effects on sediment dynamics, decomposition, and nutrient cycling in invaded lakes. As brown

bullhead catfish continue to spread across Aotearoa's rivers and lakes, management strategies aimed at protecting kōura should account for the behavioural cost of catfish NCEs alongside direct predation pressure as a, so far, underappreciated and ecologically significant dimension to this invasion.

Acknowledgements

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Authors' contributions

Study conception and design were undertaken by O.V.R, C.M. and R.H. Experiments were executed by O.V. R and C.M. Data analysis was performed by F. L. and O.V.R. Local ecological knowledge was contributed by I.A.K.K. Manuscript drafting was led by O.V.R. and C.M. All authors contributed to manuscript revision, approved the results, and the final manuscript.

Ethical approval

This study received ethical approval by University of Waikato animal ethics committee, protocol number: 1232. Additional permission to conduct experiments with kōura was granted by Te Arawa Lakes Trust and Te Komiti Whakahaere.

Declarations

The corresponding author has declared that none of the authors has any competing interests.

Data availability statement

The code and data supporting the findings of this study are openly available on Zenodo (add link) and GitHub https://github.com/OlivierRaven/NCE_Koura_catfish.

Figures

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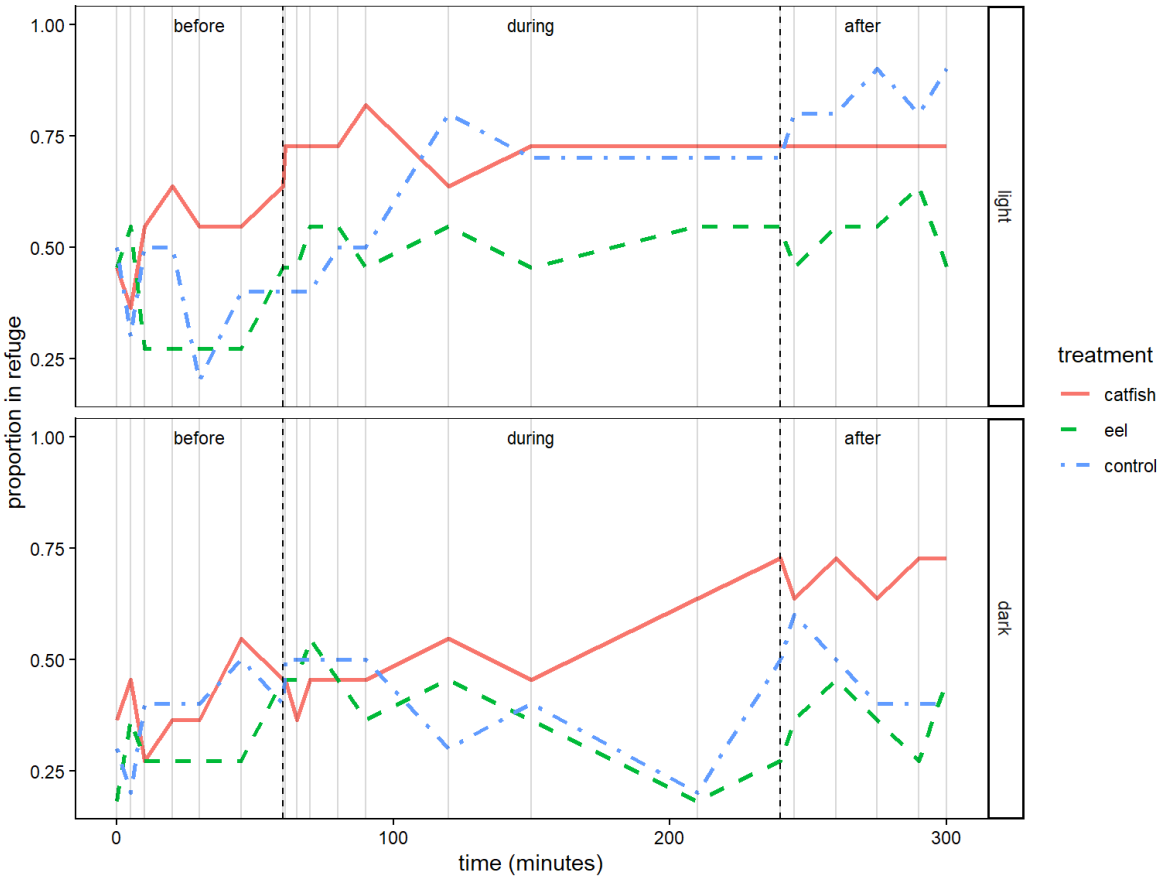

Figure 2: Kōura use of refuge over course of experiment with the three periods (before, during, after), the treatment (catfish; red-solid circle, eel; green-dashed triangle, control; blue-dotdash square), and divided over light regime (combined, dark, light). The points represent the proportion within each period for each treatment.

Source: Analysis Notebook

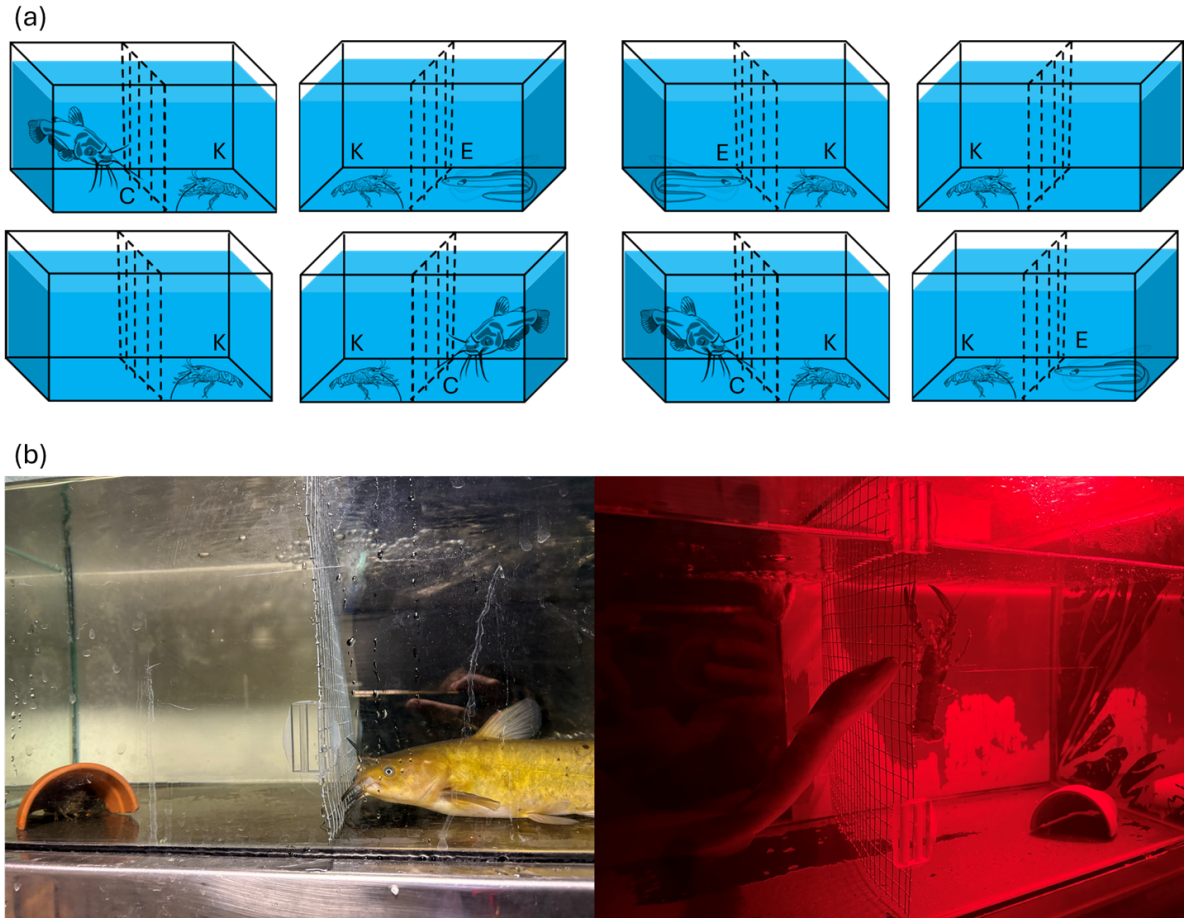

Figure 1: (a) Aquarium layout for kōura refuge seeking behaviour experiment. Each experimental round consisted of eight aquaria run simultaneously. Each tank contained a single kōura (K) and refuge (half terracotta plant pot) on one side of a mesh divider. On the other side was either a single catfish (C), eel (E) or a control (C) with no fish. The mesh divider allowed visual and chemical cues while preventing direct physical contact between tank occupants. (b) Experiments were repeated in light and dark (red light), with photographs showing a catfish in the light regime (left) and and an eel in the dark regime (red light) (right) separated from kōura by mesh dividers.

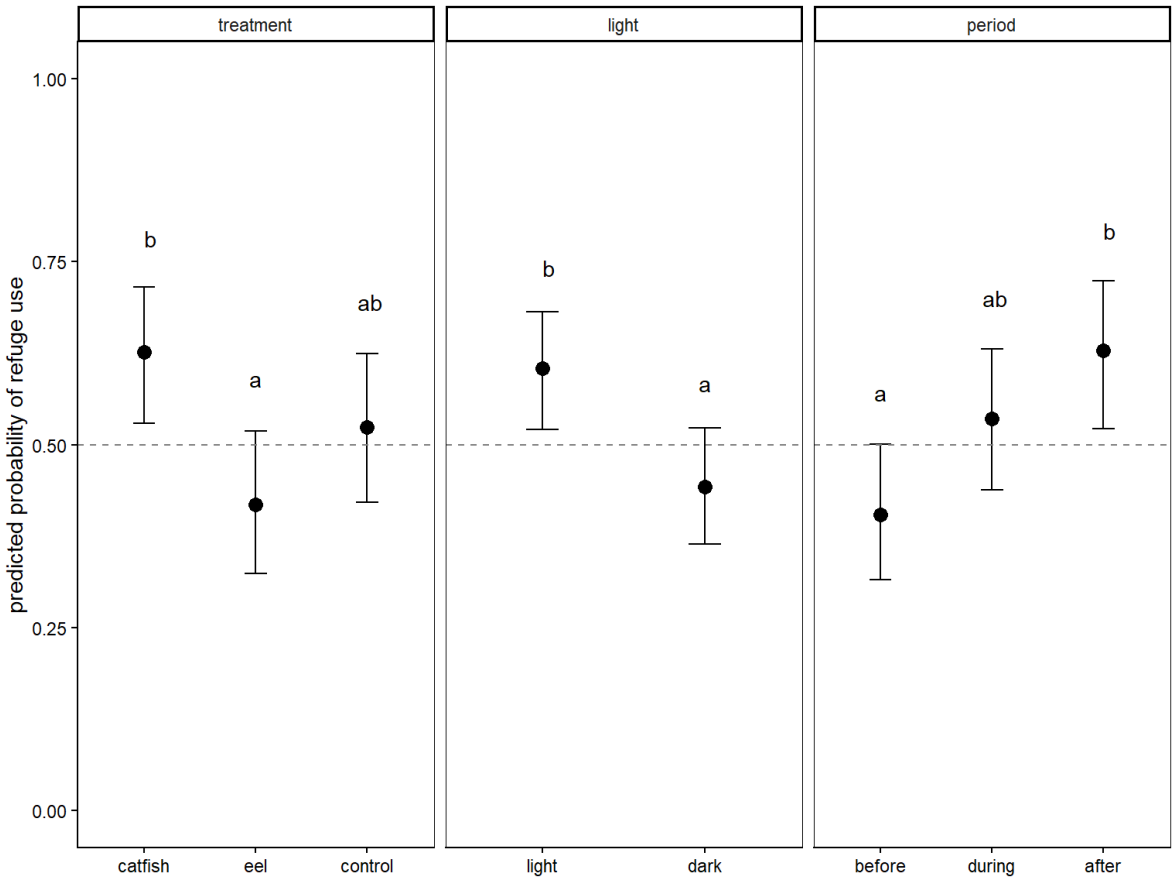

Figure 3: Predicted marginal probabilities of refuge use from the beta-binomial GLMM for each level of treatment, light condition, and experimental period. Points represent model-estimated marginal means averaged over other factors, with 95% confidence intervals shown as error bars. Different letters (a, b) indicate statistically significant differences between groups within each factor based on multiple comparison-adjusted pairwise comparisons. The dashed line at 0.5 marks the threshold where kōura spend half their time in refuge.

Source: Analysis Notebook

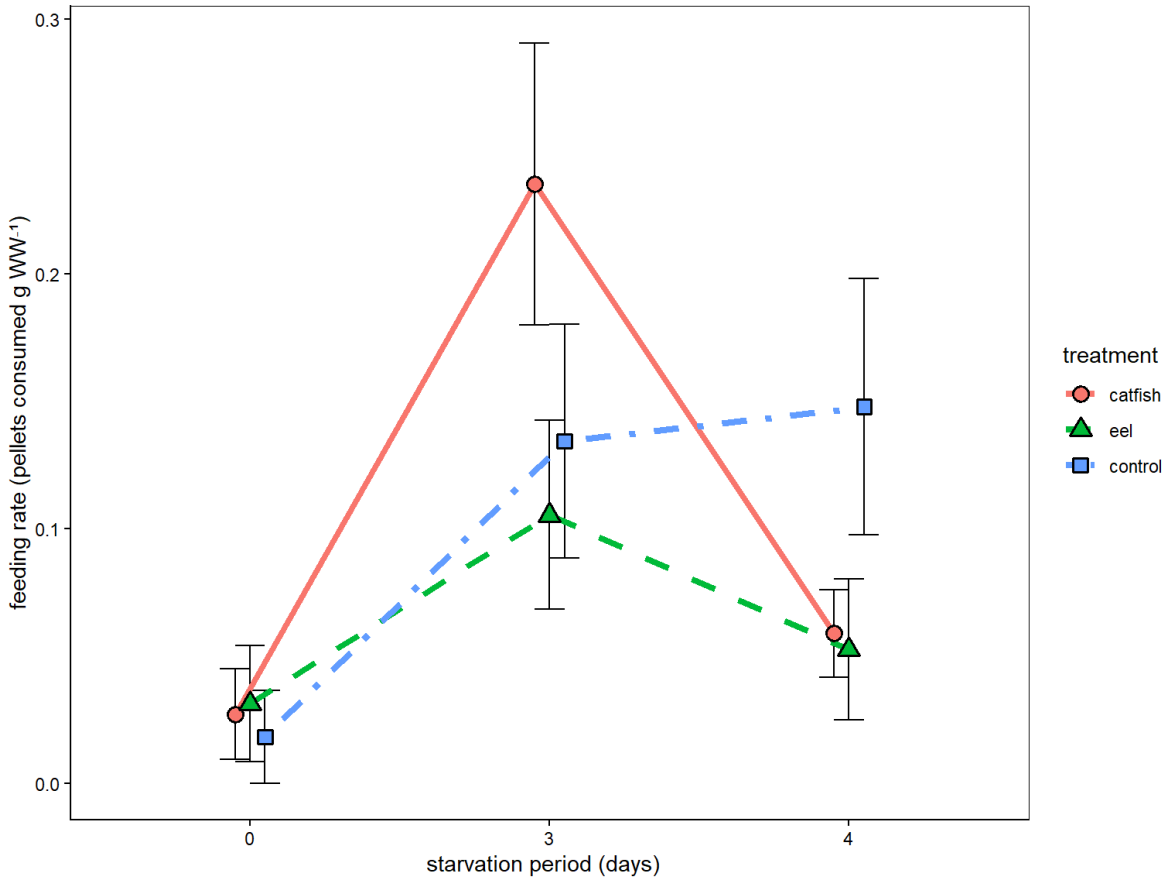

Figure 4: Mean kōura feeding rate (pellets consumed per gram body weight ± SE) across predator treatments (catfish, eel, control) and starvation periods (0, 3, and 4 days). Points represent treatment means and error bars show standard error.

Source: Analysis Notebook

Supplementary information

Source: Analysis Notebook

Table 2: Fixed effects of the beta-binomial GLMM for kōura refuge use.

Term	Estimate	Std. Error	z value	p value
(Intercept)	-0.058	0.282	-0.206	0.837
treatmentcatfish	0.424	0.292	1.452	0.146

Table 2: Fixed effects of the beta-binomial GLMM for kōura refuge use.

Term	Estimate	Std. Error	z value	p value
treatmenteel	-0.427	0.294	-1.452	0.147
lightdark	-0.654	0.239	-2.737	0.006
periodduring	0.530	0.282	1.881	0.060
periodafter	0.912	0.300	3.040	0.002

Source: Analysis Notebook

Table 3: Pairwise contrasts from the beta-binomial GLMM for kōura refuge use. Odds ratios and p-values are shown for treatment, light condition, and period comparisons. P-values are Tukey-adjusted for multiple comparisons within each factor.

Contrast	Odds ratio	SE	df	Null	z value	p value	Factor
control / catfish	0.655	0.191	Inf	1	-1.452	0.314	treatment
control / eel	1.533	0.451	Inf	1	1.452	0.314	treatment
catfish / eel	2.341	0.681	Inf	1	2.924	0.010	treatment
light / dark	1.924	0.460	Inf	1	2.737	0.006	light
before / during	0.589	0.166	Inf	1	-1.881	0.144	period
before / after	0.402	0.121	Inf	1	-3.040	0.007	period
during / after	0.682	0.205	Inf	1	-1.276	0.409	period

Source: Analysis Notebook

Table 4: Type III ANOVA table for the linear model of kōura feeding rate (pellets consumed per gram body weight). Treatment, starvation period, and their interaction were included as fixed effects.

Term	Sum Sq	Df	F value	p value
(Intercept)	0.004	1	0.640	0.429
treatment	0.000	2	0.032	0.968
day_starve	0.132	2	9.571	0.000
treatment:day_starve	0.062	4	2.241	0.082
Residuals	0.268	39	NA	NA

Source: Analysis Notebook

Table 1: Proportion of kōura observations recorded in refuge across experimental treatments (catfish, eel, control), light conditions (light, dark), and periods (before, during, and after predator exposure). Combined rows summarise across all levels of a given factor. Values represent the proportion of total observations where kōura were located in the refuge.

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